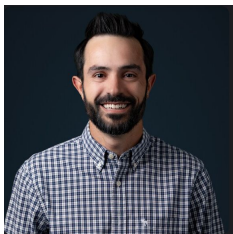


2 Hours to Data Innovation with Cloudera Lakehouse Data Engineer & Warehouse

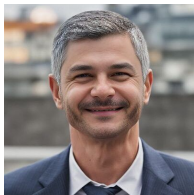
Cloudera Team



Everton Fernandes
Sr. Manager, Solutions Engineer



João Caseiro
Senior Solutions Engineer



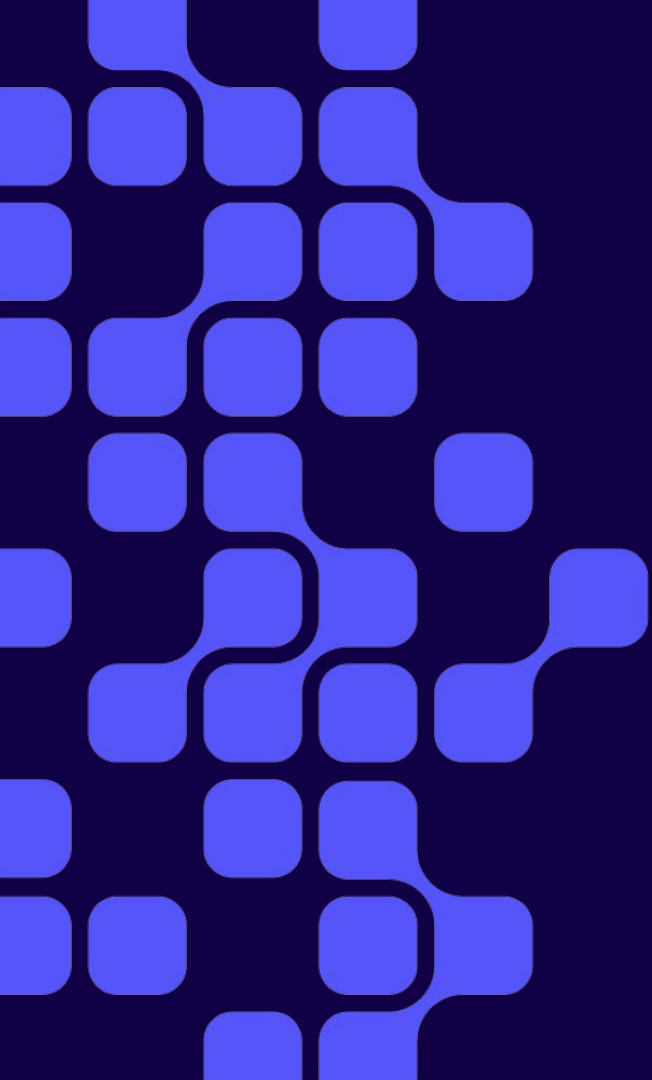
José Guimarães
Senior Solutions Engineer



Leandro Gomes
Senior Solutions Engineer



Vitor Conde
Senior Solutions Engineer



CLOUDEXERA

Open Data Lakehouse

A DEFINITION OF THE DATA LAKEHOUSE

Gartner's Perspective

A **Data Lakehouse** integrates and unifies the capabilities of a Data Warehouse and Data Lake, aiming to support AI, BI, ML and Data Engineering on a single platform.

Exploring Lakehouse architecture & use cases, Jan 2022

<https://www.gartner.com/en/documents/4010269>

Apache Hive

- Introduced by Facebook → facilitate the querying of HDFS file data using SQL, replacing the need for Java or C++ programming.

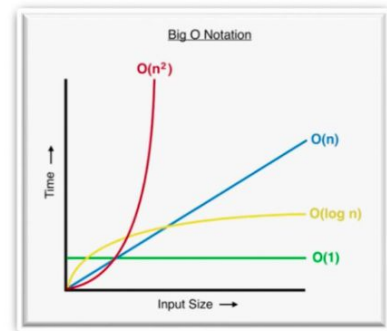


- **Hive comprises:**

- Table **metadata** and the organization of **file locations**.
- An **SQL Query engine** designed to locate data within file(s).

- **Challenges**

- No. of files increases → Hive SQL engine will search through more directories/files → **degradation of performance**.
- According to the Big O notation, Hive SQL queries exhibit an **$O(n)$ complexity** in query planning.



Hive tables

- **Pros**

- High throughput for optimized workloads
- Open data formats

- **Cons**

- No transactions
- Expensive FS operations (e.g.: list)
- Column operations are unsafe
- Partition values must be treated separately

/user/hive/warehouse/logs

```
├── dt=2001-01-01/
│   ├── country=GB/
│   │   ├── file1
│   │   └── file2
│   └── country=US/
│       └── file3
└── dt=2001-01-02/
    ├── country=GB/
    │   └── file4
    └── country=US/
        ├── file5
        └── file6
```

New Business
App/Dashboard/Report
Request

Discover &
Curate

Prepare &
Transform

Enrich (Data
Engineering)

Develop AI
Augmentation

AI-Driven Insight

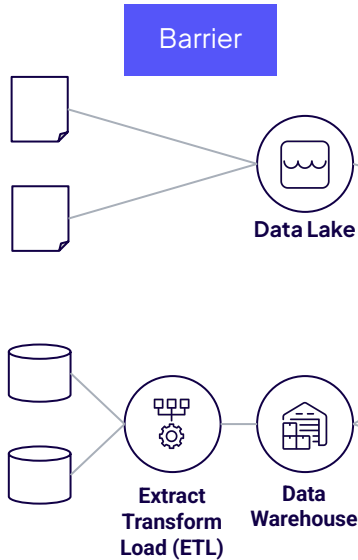
Website and
3rd Parties

Internet of
Things (IoT)

Operational
Systems

Data
Warehouses

1: Multiple Tools to Discover & Curate Data



2: Unstructured Analysis
isolated to the data lake

3: Structured Analysis
without unstructured Data

Data
Preparation

SQL

4: Augmenting Dashboards with
Prepared Data from the Lake

Data Scientist

Advanced
Analytics

Data Science
Machine Learning
Artificial Intelligence

5: Incorporating
DS/ML/AI into
real-time predictive
dashboards

6: Delivering AI/ML-driven
Augmented Reports

7: Managing a continuous loop of request to result

New App Request

ETL Expert

Data Engineer

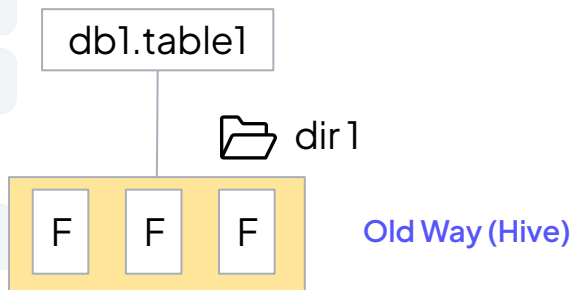
Data Analyst

Business Analyst

Consumer

How can we resolve these issues?

We need a new table format



NETFLIX 's Goals

- Table correctness/consistency
- Faster query planning and execution
- Allow users to not worry about the physical layout of the data
- Table evolution

Apache Iceberg is the Industry Standard

Who else is driving the community forward?



CLUDERA



NETFLIX

Tencent



Started by **Netflix** and became a top-level **Apache** project in May 2020

Active **open source community** with growing set of **compute engine** support

Strong **industry adoption** & recognized as **next-gen cloud native** table format

Ease-of-use & **enhanced features** for PB scale datasets

We are an **active contributor** of Iceberg

Table formats

Abstracting storage from compute, Data warehouse management in the data lake



Creator

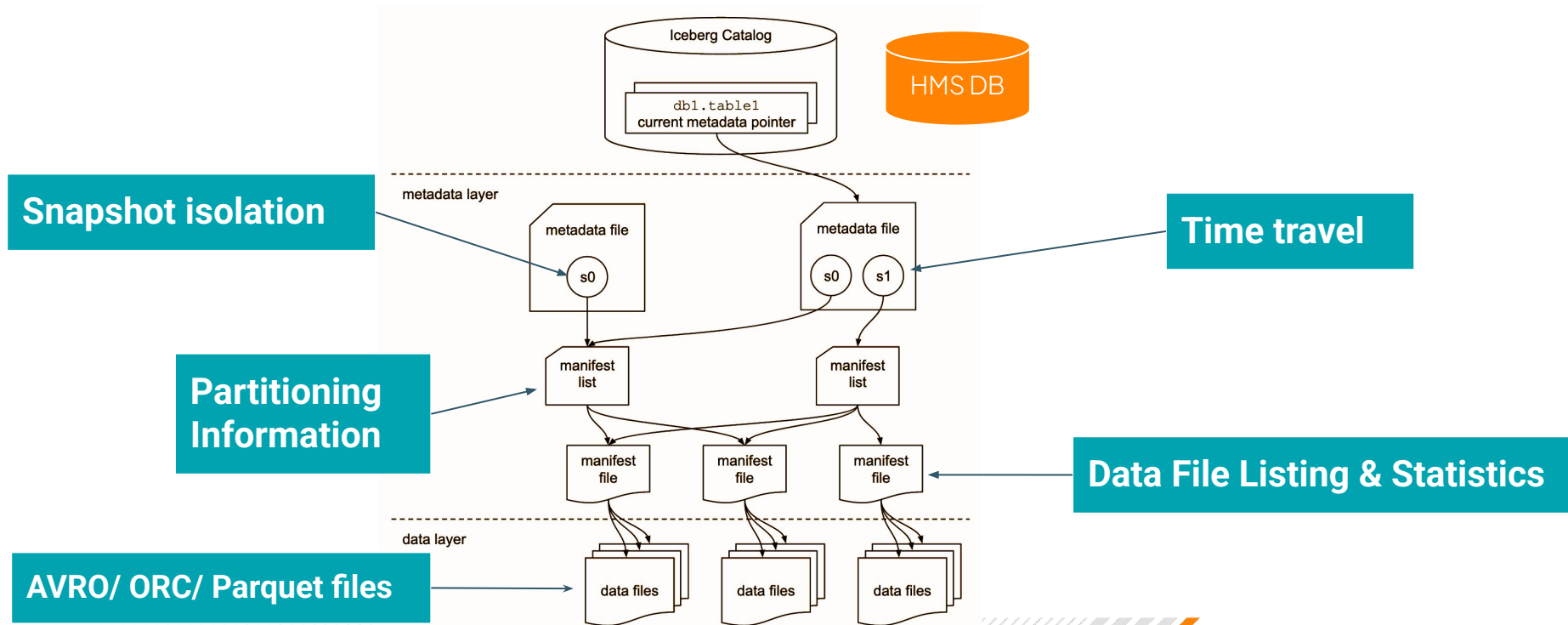
Uber



NETFLIX

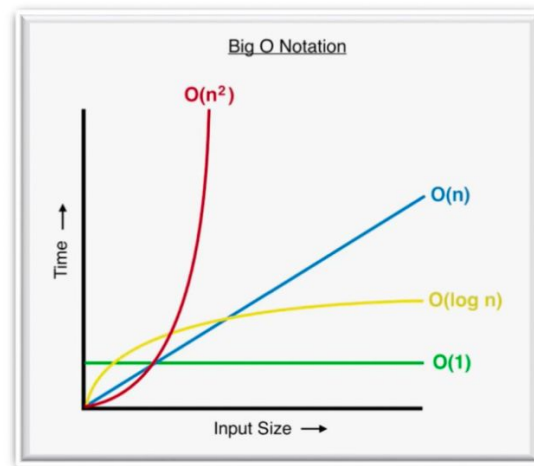
CLOUDERA

How does Iceberg Format solve the Issues



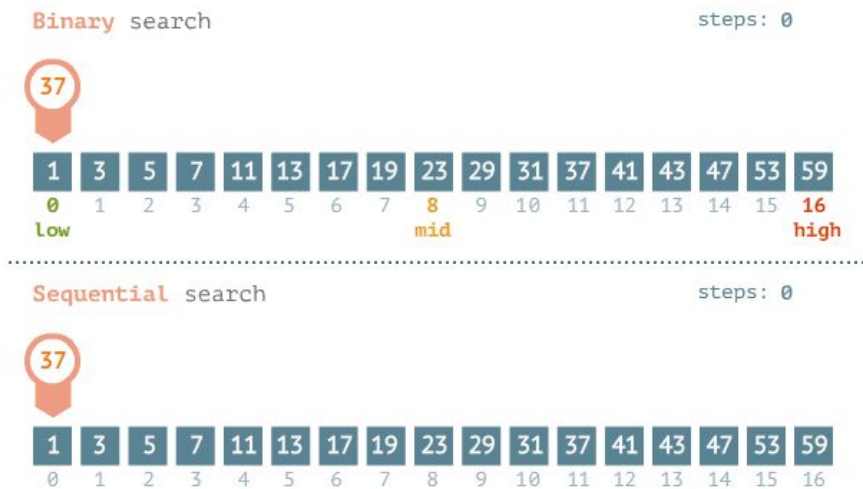
SQL Performance on Iceberg table format: CONSISTENT SLA

- Iceberg tables consist of a list of files.
 - These tables allow direct access to files through manifest lists and manifest files.
 - Following the Big O concept, Iceberg queries exhibit an **$O(1)$ complexity** in query planning.
 - **Regardless of the table's size**, the query access plan remains consistently constant within the Iceberg approach.



Big O notation

The moon landing - How fast is your application?



www.mathwarehouse.com



SQL Performance on Iceberg table format: CONSISTENT SLA

No performance degradation when running queries on 1B or 10B records

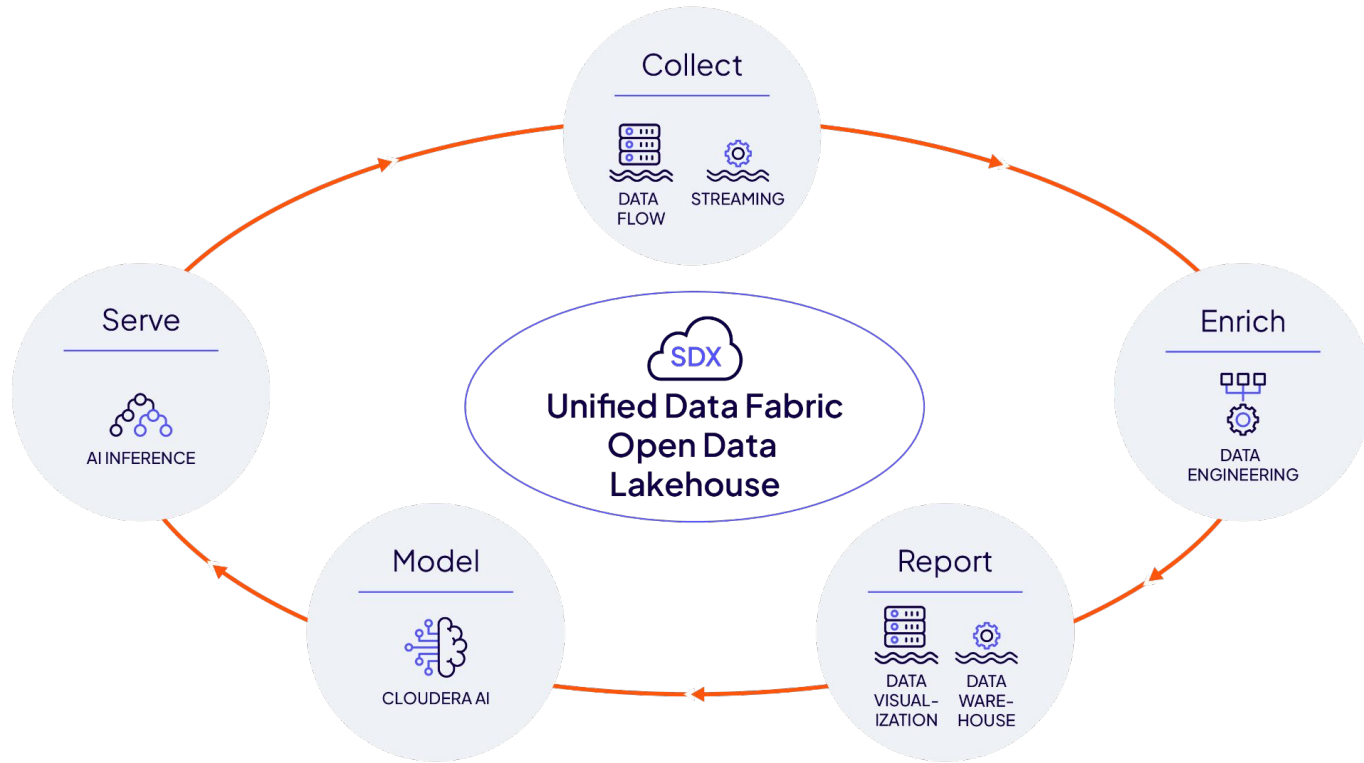
Analyzed 100+ Complex SQL Statements, see the following example.

Reduced Load on YARN: almost -40%

Row Count	Query Time	Comment
1.7 Billion	2 secs	
5.2 Billion	2.2 secs	
10 Billion	9 sec	Updates, Deletes, and Data Skew have impacted SQL Performance
10 Billion	2.5 sec	After Table maintenance

Data Lifecycle

The industry's best enterprise-grade utilities integrated on one data management platform.



New Business
App/Dashboard/Report
Request

Discover &
Curate

Prepare &
Transform

Enrich (Data
Engineering)

Develop AI
Augmentation

AI-Driven Insight

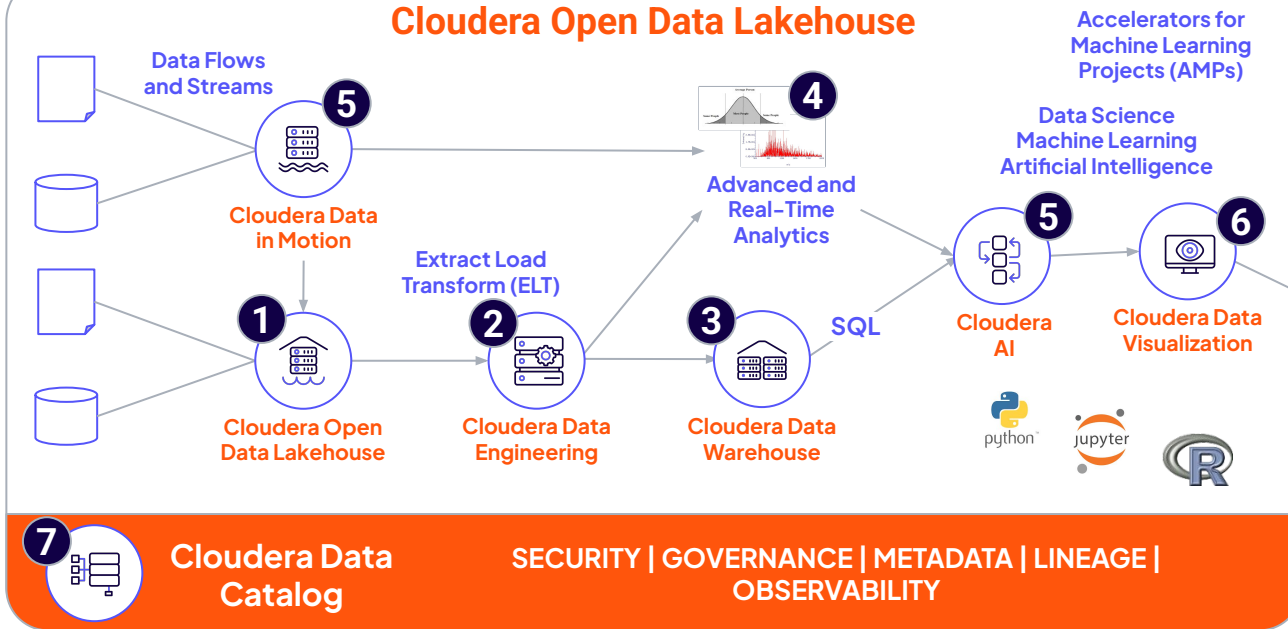
Website and
3rd Parties

Internet of
Things (IoT)

Operational
Systems

Data
Warehouses

Cloudera Open Data Lakehouse



New App Request

ETL Expert

Data Engineer

Data Analyst

Business Analyst

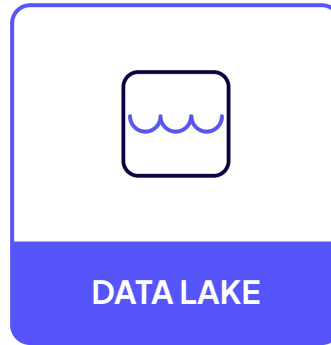
Consumer

Data Lakehouse: Data Lake + Warehouses

Accelerate time to insights while reducing analytic costs



Structured
Well curated data
Precise analytics
Built for SQL
Well governed
Inflexible & \$\$\$



Semi-structured &
Unstructured data
AI, ML, & Predictive
Analytics
**Poor Data quality &
Slow BI queries**



BEST of Both Worlds

Unified View
of Data

Flexibility
and choice

No Vendor
Lock-in

CLOUDERA

DATA LAKEHOUSE - KEY CONSIDERATIONS

Unified data

- Common metadata store for raw and curated data
- Common security, governance & lineage
- Integrated data engg for streamlined data curation

Flexible

- Best in class performance with multiple engines
- Multiple file formats
- Support for batch and streaming data

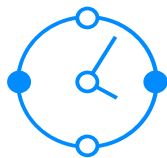
Open

- Open table format
- Open ecosystem
- Open source
- Open standards
- Community adoption
- No vendor lock-in
- Hybrid, multi-cloud

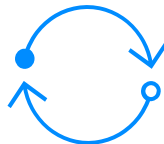
KEY FEATURES



ACID
Transaction



Time Travel /
Table Rollback



In-place
Table
Evolution

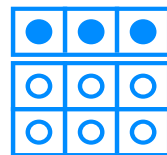
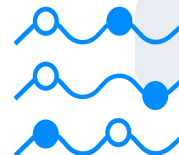
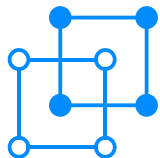


Table
Maintenance



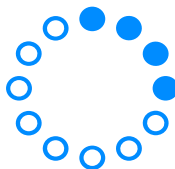
Streaming &
Batch Ingestion



Iceberg
Replication



SDX
Integration



Ease of
Adoption



Multi-Hybrid
Cloud

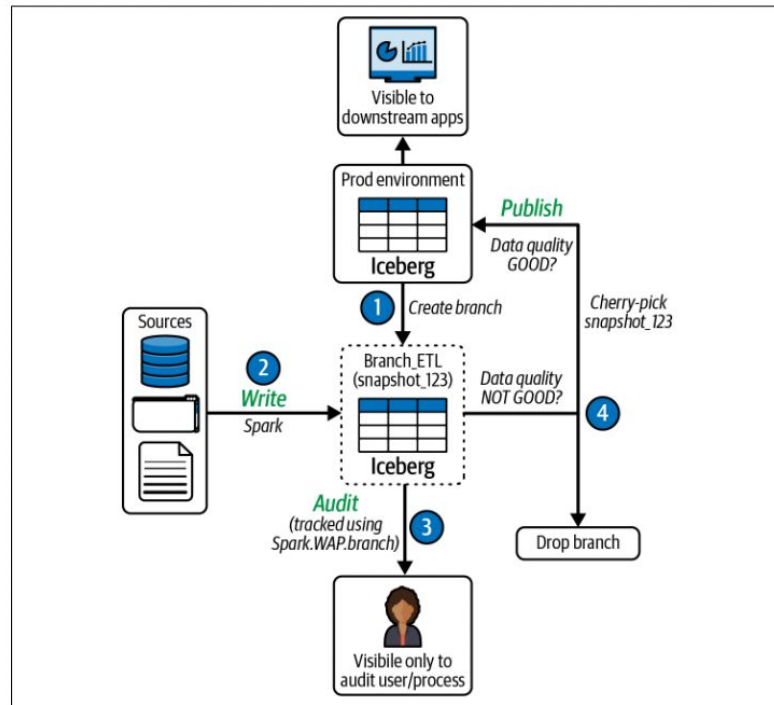


Performance
/ Scalability

CLUDERA

WAP Using Iceberg's Branching Feature

The WAP process at the table level



Cloudera Lakehouse Advantage

Ease of Use , Shorter time to modern, advanced use cases and reduced costs



Increase
In Data Engineer
Productivity



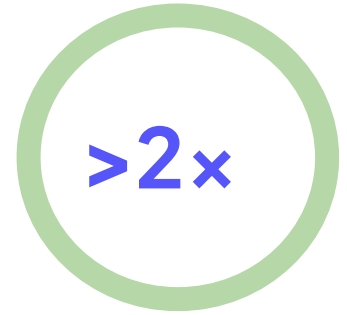
Increase
In Data Analyst
Productivity



Increase
In Data Scientist
Productivity



Improvement In
Compute Efficiencies



Reduction In Storage
Costs

Sources: Anaconda, Wakefield Research, IDC, Experian

Note: Observations from our customers and in-house implementations

CLOUDERA

Data Engineering

Data Engineering

“É o desenvolvimento, a implementação e a manutenção de sistemas e processos que recebem dados brutos e produzem informações consistentes e de alta qualidade que dão suporte a casos de uso posteriores, como análise e aprendizado de máquina”

R. Buckminster Fuller

Data Engineering $\neq$



It involves multiple capabilities...



Processing
Engine



Pipelines
Orchestration



Monitoring &
Debugging



Automation &
Operation

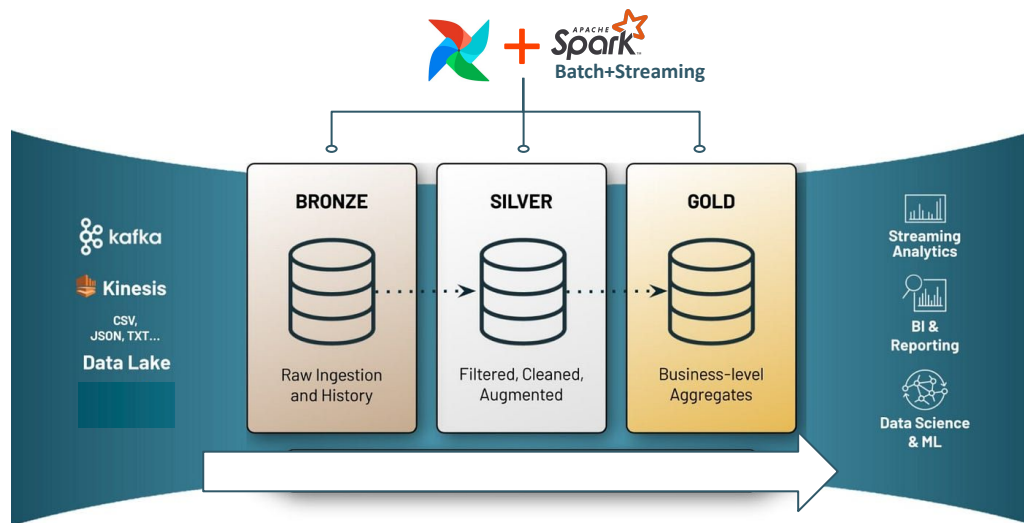
Data Engineering is a core component of an Open Lakehouse

Simplify Development

Build complex pipelines with ease

Effortless diagnosis & optimization

Combine batch & streaming



Cloudera Data Engineering (CDE)

An integrated, purpose built experience for Data Engineers

Containerized, Managed Spark Service



Autoscaling compute
Governed & secure with
Cloudera SDX
Mix version deployments
K8s optimized resource
scheduling (Yunikorn)

Apache Airflow Orchestration



Open preferred tooling
Orchestrate complex data
pipelines
Manage & schedule
dependencies easily

Tuning & visual Troubleshooting



Resolve issues fast with
real-time visual performance
profiling
Complete monitoring & alerting
capabilities

Simplified Jobs Management & APIs

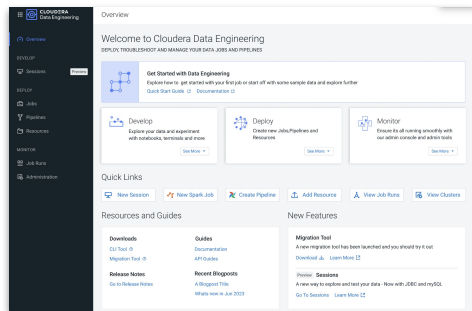


Full lifecycle management
API-driven pipeline automation
for any service
Any language: SQL, Java, Scala,
Python

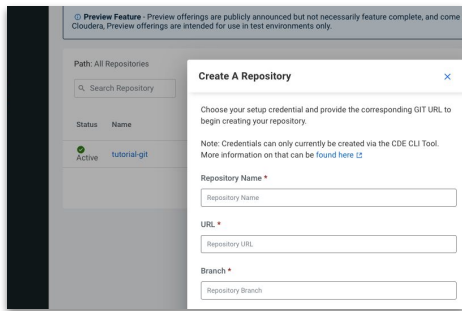
Some Features

USERS

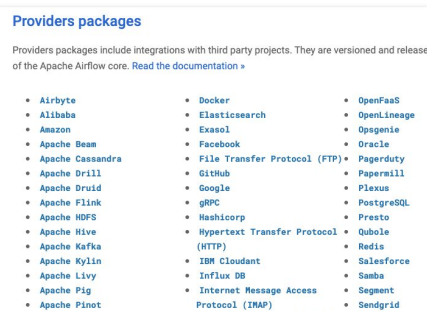
Refreshed Home



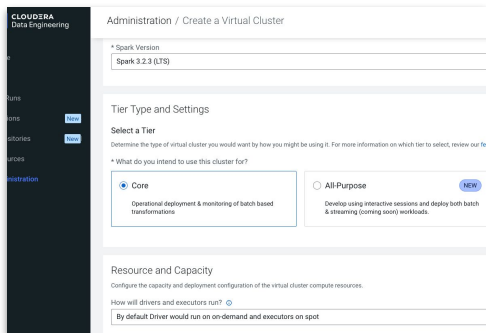
Git Repositories



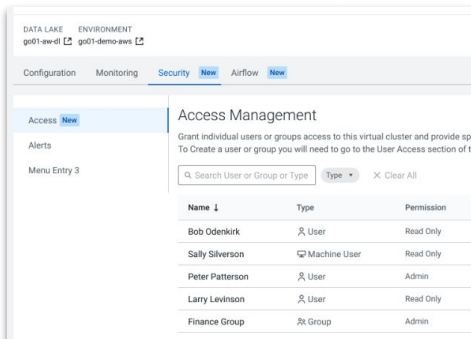
Airflow custom operators



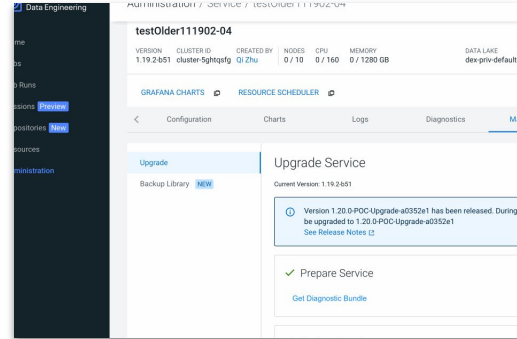
Tiered VCs



Group ACLs & Roles



In-place Upgrade



ADMINS

CLOUDERA

Data Warehouse

Cloudera Data Warehouse

Cloudera differentiated value for Practitioners

The screenshot shows the Cloudera Data Warehouse Service Overview page. It features a sidebar with navigation icons, a main header with 'Overview', and a welcome message. Below the welcome message, there are three main sections: 'Get Started With Data Warehouse' with links to 'Start Guide', 'Cloudera Patterns', 'Explore Use Cases', 'Community Forum', and 'Documentation'; 'Create' with a 'See More' button; and 'Query and Visualize Data' with a 'See More' button. There is also a 'Resources and Downloads' section with a 'See More' button. At the bottom, there is a table of virtual warehouses with columns for Status, Name, Type, Version, CPU, Executor, Apps, Uptime, and Actions. Two virtual warehouses are listed: 'Application-1' and 'Line-of-Business-1', both with a status of 'Good Health' and a type of 'Trino'.

Overview

Welcome to Cloudera Data Warehouse Service

Cloudera Data Warehouse (CDW) is a cloud-native self-service analytic experience that enables BI analysts to go from zero to query in minutes.

Get Started With Data Warehouse

These resources can help you to learn how to use Cloudera Data Warehouse.

[Start Guide](#) [Cloudera Patterns](#) [Explore Use Cases](#) [Community Forum](#) [Documentation](#)

Create

Create new environments, database catalogs, virtual warehouses

[See More](#)

Query and Visualize Data

Run SQL queries and create reports, or other visualizations you can share

[See More](#)

Resources and Downloads

Documentation, release notes, JDBC/ODBC drivers, CLI client downloads, UDF SDKs, and more

[See More](#)

Environments (83) Database Catalogs (12) **Virtual Warehouses (20)**

Search by id or name Status Type Environment Database Catalog X Clear All [New Virtual Warehouse](#)

Status	Name	Type	Version	CPU	Executor	Apps	Uptime	Actions
Good Health	Application-1 trino-1753300184-ddhz dw-learn-env-cl-default dw-learn-env	Trino	2025.0.20.0-191	42		HUE	26 minutes	Suspend
Good Health	Line-of-Business-1 trino-1753300159-mnsv dw-learn-env-cl-default dw-learn-env	Trino	2025.0.20.0-191	42		HUE	27 minutes	Suspend

Focused experience

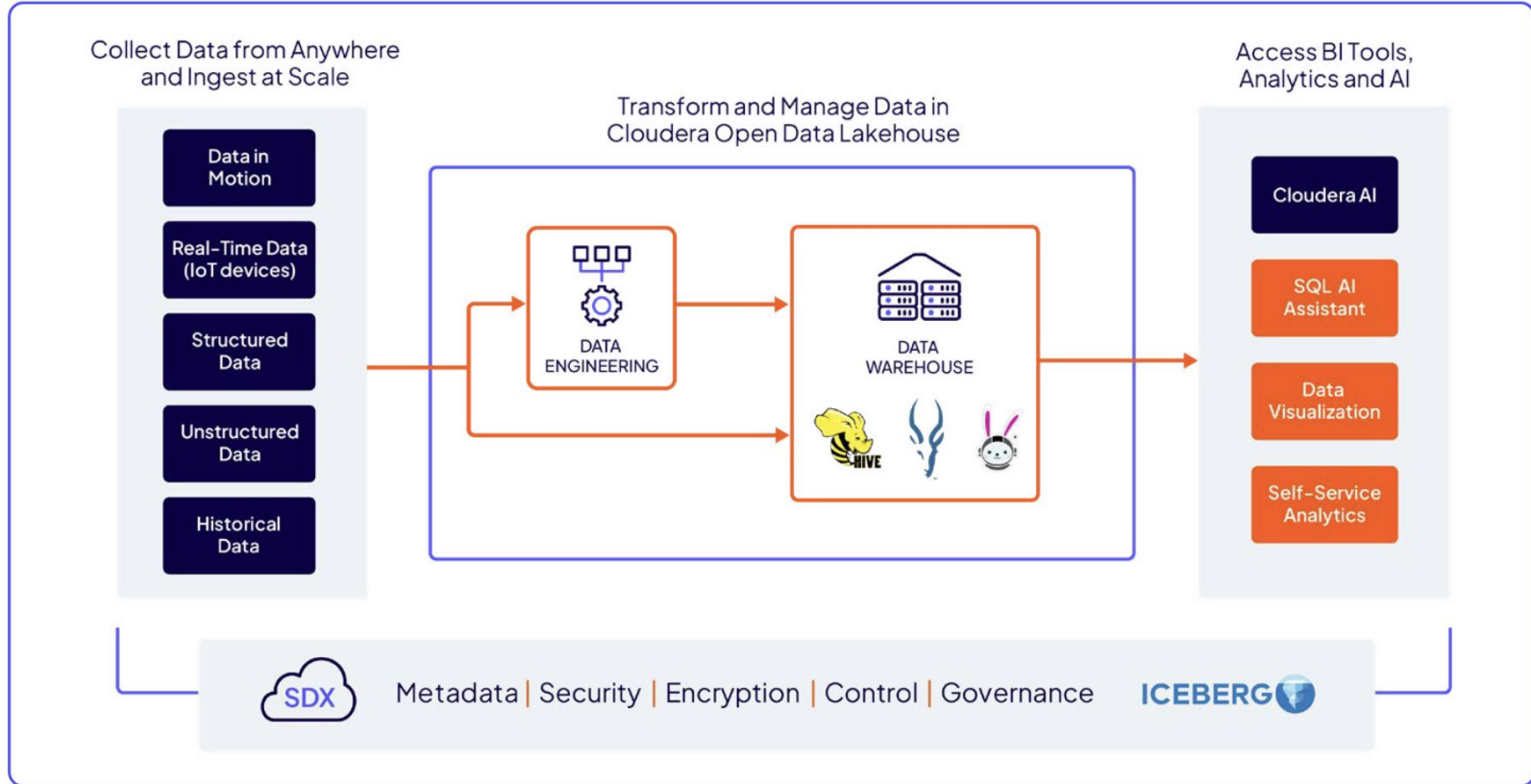
- Any data, Anywhere
- Guided virtual warehouse provisioning
- SQL syntax generator
- Natural language / talk-to-your-data
- Integrated Analytics Ecosystem

Virtual Warehouse operation

- Auto scaling
- Auto start/resume
- Auto stop/pause

A Comprehensive Data Platform

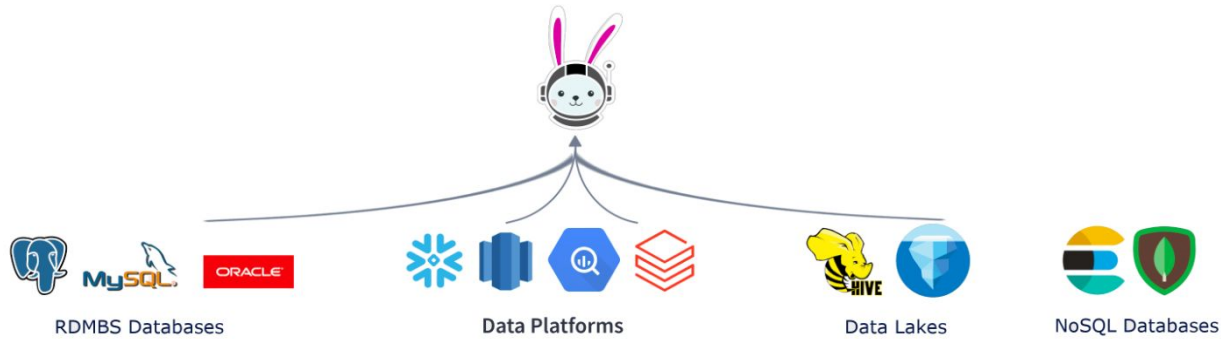
Cloudera Data Warehouse is a key offering in the Cloudera Open Data Lakehouse architecture



Cloudera Data Warehouse – Trino

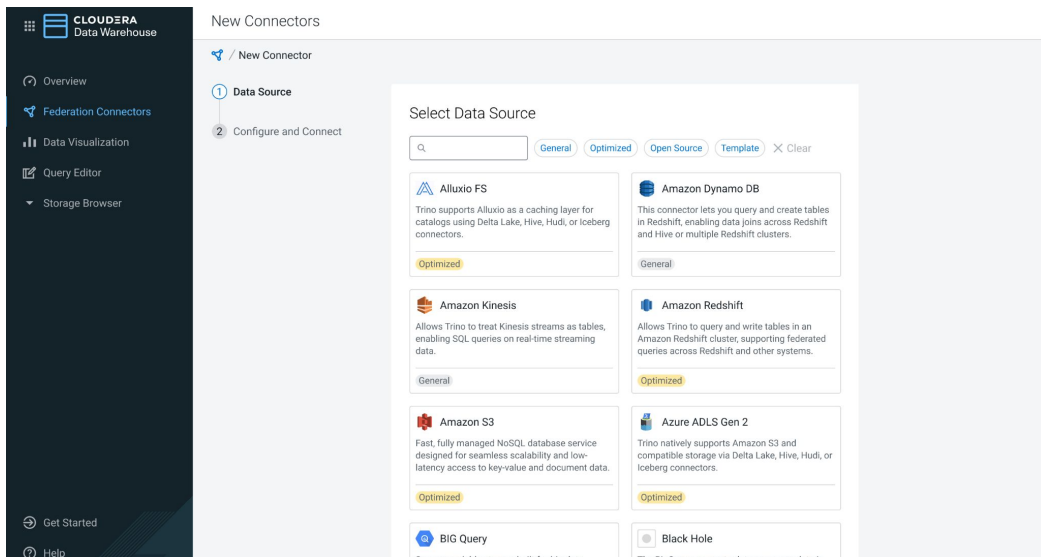
What is Trino?

Trino is an open-source distributed SQL query engine that's used for big data analytics and data engineering. Trino is used to run interactive queries and batch processing on large amounts of data. It can query data from multiple data sources, including data lakes, and run federated queries across different data sources.



Cloudera Data Warehouse – Trino

Cloudera differentiated value for Administrators



Data security & governance

- Inherit platform-wide governance policies
- Resource-based Authorization, Column Masking, Row Filtering, and Data Lineage

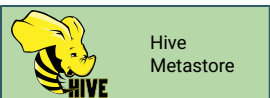
Ecosystem configuration

- Guided federation connector setup
- Subscription model across Virtual Warehouses
- True hybrid deployment

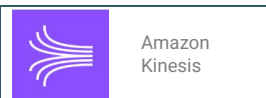
Cloudera Data Warehouse – Trino

Cloudera Trino: Connectors

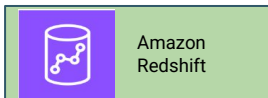
Cloudera Optimized Connectors



Hive
Metastore



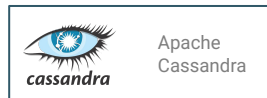
Amazon
Kinesis



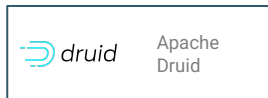
Amazon
Redshift



Apache
Accumulo



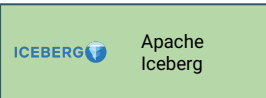
Apache
Cassandra



Apache
Druid



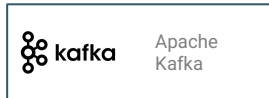
Apache
Hudi



Apache
Iceberg



Apache
Ignite



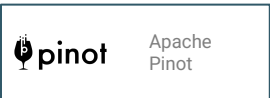
Apache
Kafka



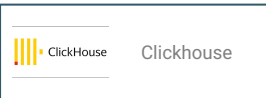
Apache
Kudu



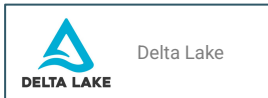
Apache
Phoenix



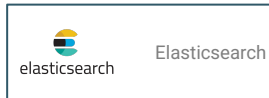
Apache
Pinot



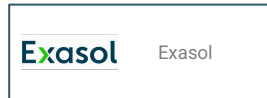
Clickhouse



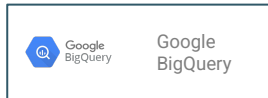
Delta Lake



Elasticsearch



Exasol



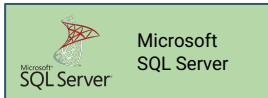
Google
BigQuery



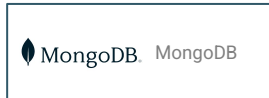
Vast



MariaDB



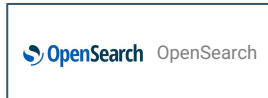
Microsoft
SQL Server



MongoDB



MySQL



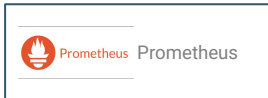
OpenSearch



Oracle



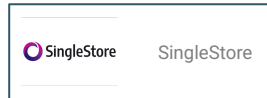
PostgreSQL



Prometheus



Redis



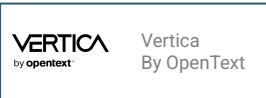
SingleStore



Snowflake



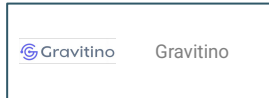
TPC



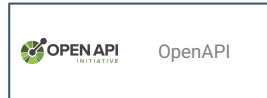
Vertica
By OpenText



Git



Gravitino



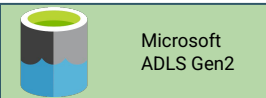
OpenAPI



Teradata



Amazon
S3



Microsoft
ADLS Gen2



Google
GCS



Apache
HDFS



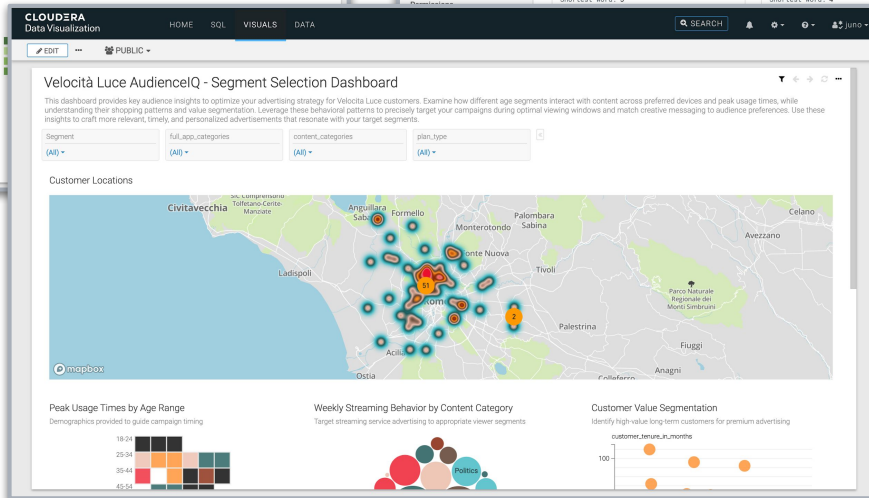
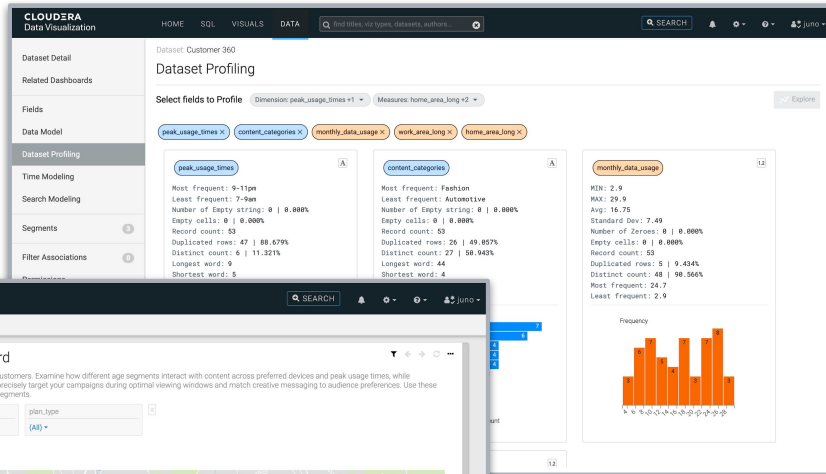
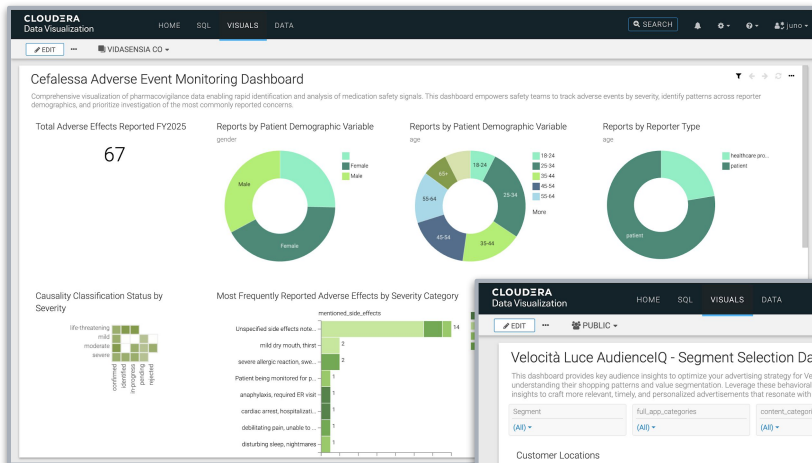
Alluxio
FS



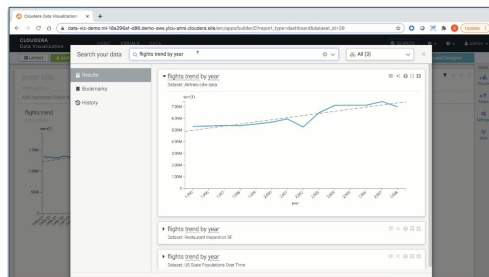
local
FS

Cloudera Data Visualization

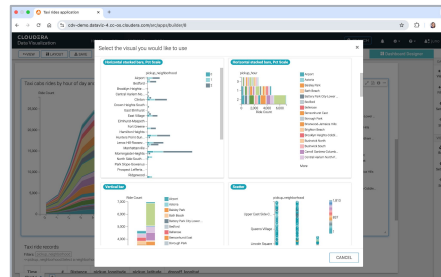
Integrated visualization and dashboarding with Cloudera Data Warehouse



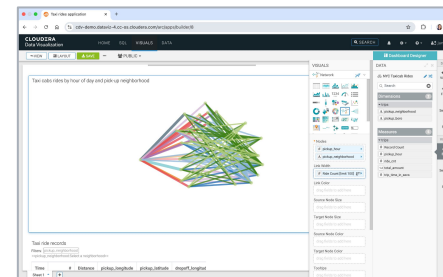
Fully Featured Visual App Building Environment



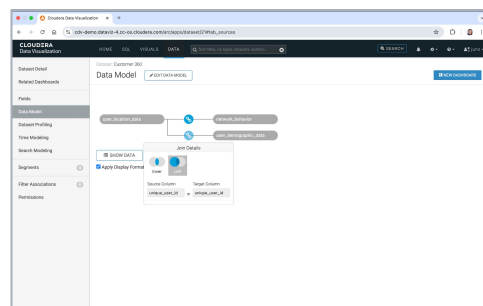
Natural language search



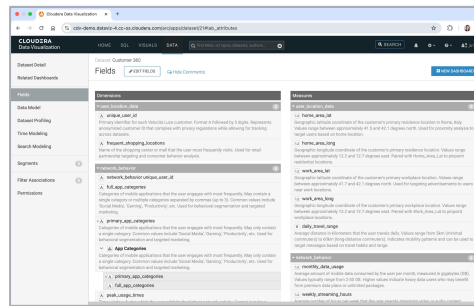
Visual recommendations



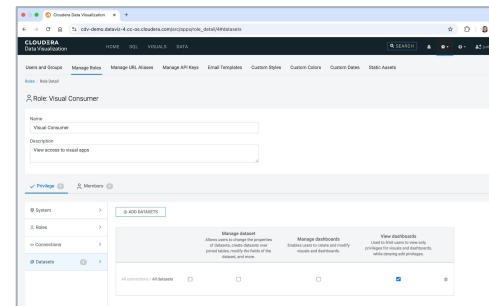
Email integration, alerting & job scheduling



Blend data easily
joins, hierarchies, geo types



Build & share a logical semantic layer



Role based security enables governance
and access controls

CLOUDERA

Demo

Overview

Welcome to Cloudera Data Warehouse Service

Cloudera Data Warehouse (CDW) is a cloud-native self-service analytic experience that enables BI analysts to go from zero to query in minutes.



Get Started With Data Warehouse

These resources can help you to learn how to use Cloudera Data Warehouse.

[Start Guide](#)



Create

Create new environments, database catalogs, virtual warehouses

[See More](#)



Query and Visualize Data

Run SQL queries and create reports, or other visualizations you can share

[See More](#)



Resources and Downloads

Documentation, release notes, JDBC/ODBC drivers, CLI client downloads, UDF SDKs, and more

[See More](#)

Environments (1)

Database Catalogs (1)

Virtual Warehouses (0)

Search by id or name

Status

Environment

Clear All

[New Database Catalog](#)

Status	Name	Virtual Warehouses	Version	Uptime	Actions
Good Health	host-1 warehouse-1753261324-q558 host-1	0	2025.0.19.4-5	5 days	Suspend

Hands On Lab

Hands On Lab

Módulo do dia

Módulos

- 01 - CDE
- 02 - Hive
- 03 - Impala

Planilha de usuários

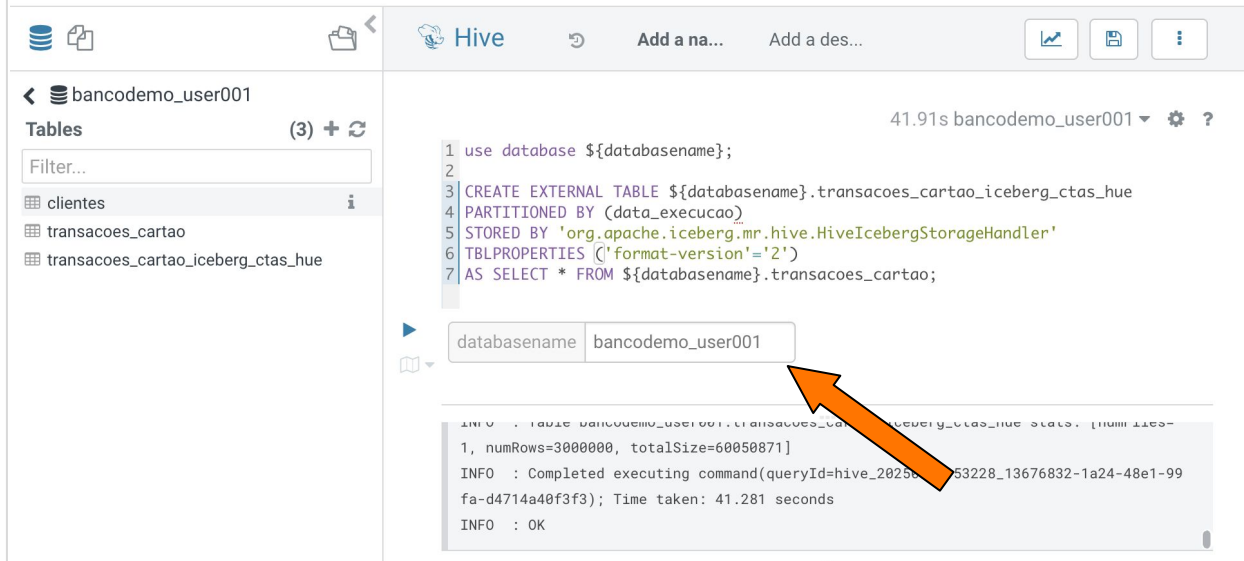
<https://shorturl.at/9fRXK>



HUE - SQL

Vamos usar bastante o Hue

Vamos usar alguns variáveis
para ajudar “**variáveis**”



The screenshot displays the Hue SQL interface. On the left sidebar, the user is logged in as 'bancodemo_user001' and the 'Tables' section shows three tables: 'clientes', 'transacoes_cartao', and 'transacoes_cartao_iceberg_ctas_hue'. The main panel shows a Hive query using variables: `use database ${databasename};`, `CREATE EXTERNAL TABLE ${databasename}.transacoes_cartao_iceberg_ctas_hue PARTITIONED BY (data_execucao_...)`, `STORED BY 'org.apache.iceberg.mr.hive.HiveIcebergStorageHandler'`, `TBLPROPERTIES ('format-version'='2')`, and `AS SELECT * FROM ${databasename}.transacoes_cartao;`. Below the query, a variable input field for 'databasename' is set to 'bancodemo_user001'. An orange arrow points to this field. The bottom panel shows the execution output: 'INFO : Table bancodemo_user001.transacoes_cartao_iceberg_ctas_hue stats: (num files=1, numRows=3000000, totalSize=60050871)', 'INFO : Completed executing command(queryId=hive_2025053228_13676832-1a24-48e1-99fa-d4714a40f3f3); Time taken: 41.281 seconds', and 'INFO : OK'.

Workshop Environment

Access to Environment

FIELD-MARKETING-AMER

Sign in to your account

Username or email

user099

Password

Sign In

Access CDE

The screenshot displays the Cloudera Data Platform (CDP) interface. At the top, the Cloudera logo and 'Data Platform' text are visible on the left, and a search bar with the text 'How can we help you?' is in the center. On the right, there are links for 'Enable New UI' and the user profile 'Jessica Oliveira'. Below the header, a welcome message says 'Welcome, Jessica. What would you like to work on?'. A 'Favorites' section is present, followed by an 'All Services' section. In the 'All Services' section, the 'Data Engineering' service is highlighted with a red box and a red arrow pointing to it. Below this, the 'Observability Analytics Summary' section shows activity for 'Data Hub Clusters', 'Data Engineering', and 'Data Warehouse'. The 'Data Engineering' section shows activity between 10/22/2024 and 10/24/2024, with 1 Failed, 0 Slow, and 3 Total Spark Jobs. The 'Data Warehouse' section shows activity between 10/23/2024 and 10/25/2024, with 133 Failed, 149 Slow, and 1073 Total Impala Queries.

CLUSTERA
Data Platform

How can we help you?

Enable New UI Jessica Oliveira

Welcome, Jessica. What would you like to work on?

▼ Favorites

☆ Your favorites will appear here. [Learn More](#)

▼ All Services

DataFlow Data Engineering Data Warehouse Operational Database Machine Learning Data Hub Clusters Data Catalog Replication Manager Observability Management Console

Observability Analytics Summary

▼ Show / Hide

Data Hub Clusters cdp-hms-spark-dh

Activity on (10/24/2024)

Impala Queries 0 Failed 0 Slow 4 Total

Data Engineering cdp-iceberg-vc

Activity between (10/22/2024 - 10/24/2024)

Spark Jobs 1 Failed 0 Slow 3 Total

Data Warehouse cdp-hol-impala-vw5

Activity between (10/23/2024 - 10/25/2024)

Impala Queries 133 Failed 149 Slow 1073 Total

CDE

The screenshot shows the Cloudera Data Engineering (CDE) Home dashboard. On the left is a dark sidebar with navigation links: Home, Jobs, Job Runs, Sessions (with a 'Preview' badge), Resources, and Administration. The main content area is titled 'Home' and 'Welcome, User99'. It features three primary action cards: 'Create' (for jobs, sessions, and ad-hoc runs), 'Resources' (for file and python resources), and 'Docs & Downloads' (for references and downloads). Below these is a 'Virtual Clusters' section showing an autoscaling Spark cluster named 'cde-innovation-srv' with a table of its current state.

CLUSTER CLOUDERA Data Engineering

Home Jobs Job Runs Sessions **Preview** Resources Administration

Home

Welcome, User99

Create
Create jobs, orchestrate them or start a session.

- Spark Jobs
Create New
Schedule
Ad-Hoc Run
- Airflow Pipelines
Upload DAG file
Build a Pipeline **New**
- Sessions **New**
Start New

Resources
Create resources for jobs.

- File
Create New
- Python
Create New

Docs & Downloads
CDE documentation and tools.

- References
API Doc
Product Doc
Release Notes
- Downloads
CLI Client
Migration Tool **New**

Virtual Clusters
Autoscaling Spark clusters to run Jobs.

cde-innovation-srv

vc-lambda [View Jobs →](#)

Spark 3.2.3

CPU	MEMORY	JOBS
0	0 MB	0

Help
User99 Test99
1.19.4-h1-b1

THANK YOU

CLOUDERA